

SHIRIN NASR-ESFAHANI

3D Reconstruction | Computer Vision & Machine Learning

nasresfah@gmail.com | (702) 666-5199 | [LinkedIn](#) | [Google Scholar](#) | [GitHub](#)

VISA STATUS: U.S. Permanent Resident (Green Card Holder)

PROFESSIONAL SUMMARY

I am a 3D computer vision and machine learning engineer with a strong focus on multi-view geometry, sensor calibration, and spatial estimation. My work centers on building accurate and scalable perception systems that extract reliable structure and motion from multimodal data. I have hands-on experience developing end-to-end reconstruction and calibration pipelines in Python and C++, using frameworks such as PyTorch and TensorFlow. My background combines classical geometric methods—camera modeling, pose estimation, and bundle adjustment—with learning-based approaches that improve robustness and generalization. I regularly work with RGB, LiDAR, thermal, and IMU data in UAV and field settings, where consistency and precision are essential. I enjoy translating research ideas into practical systems, contributing across the full pipeline from data processing and algorithm design to evaluation and deployment.

TECHNICAL SKILLS

Programming & Frameworks

- **Languages:** Python, C++, MATLAB
- **Machine Learning & Deep Learning:** PyTorch, TensorFlow, Keras, scikit-learn
- **Scientific Computing & Data Processing:** NumPy, SciPy

3D Vision & Geometry

- Multi-view geometry, camera calibration, pose estimation, and 3D scene representation
- OpenCV, Open3D, Trimesh, PyMeshLab, SciPy, scikit-image

Neural Reconstruction & Rendering

- Structure-from-Motion (SfM) and large-scale 3D reconstruction
- Neural scene representation methods (NeRF, 3D Gaussian Splatting)
- Point cloud and mesh processing for scene modeling
- Agisoft Metashape, RealityCapture, Nerfstudio, CloudCompare, Blender

Machine Learning

- CNNs, RNNs, GANs
- Reinforcement learning (Soft Actor-Critic) for adaptive calibration and spatial optimization

Perception & Sensor Fusion

- RGB, LiDAR, thermal, hyperspectral, and multispectral data
 - Spatial alignment, temporal synchronization, and feature-level fusion
 - UAV and lab platforms (e.g., Resonon Pika II, Headwall Nano HP)
-

PROFESSIONAL EXPERIENCE

Postdoctoral Researcher

Center for Intelligent Infrastructure, Missouri University of Science and Technology — Rolla, MO

May 2025 – Present | NSF-Funded Project

- Designed and implemented scalable end-to-end 3D reconstruction pipelines to generate georeferenced digital twins from UAV imagery and LiDAR data.
- Integrated multimodal sensor data (RGB, LiDAR, thermal) to improve geometric consistency and scene-level accuracy in large-scale environments.
- Developed robust 3D perception and world modeling workflows to enable spatial analysis and quantitative assessment in real-world deployments.

Postdoctoral Fellow

Electrical and Computer Engineering Department, Missouri University of Science and Technology — Rolla, MO

October 2022 - April 2025 | Project funded by the U.S. Army Research Laboratory

- Designed and implemented 3D vision algorithms for spatial estimation in GPS-denied and low-visibility conditions, combining feature-based matching with Siamese neural networks.
- Developed a reinforcement learning framework (Soft Actor-Critic) for adaptive camera self-calibration, enabling intrinsic parameter estimation and improved perception without external positioning systems.
- Built and evaluated monocular spatial estimation models under varying environmental conditions, applying Elastic Weight Consolidation and error-driven learning to enhance robustness and generalization.

Graduate Research & Teaching Assistant

Electrical and Computer Engineering Department, University of Nevada-Las Vegas (UNLV) — Las Vegas, NV

September 2017 – August 2022

- Built and curated a hyperspectral imaging dataset (400–900 nm, Resonon Pika II) with over 1,700 samples collected across multiple time intervals to support data-driven material analysis.
- Developed multimodal learning models integrating hyperspectral and RGB imagery, significantly outperforming RGB-only baselines (MATLAB, Python, TensorFlow).
- Designed and implemented a 1D CNN for defect detection and visual classification tasks.
- Supported instruction in Digital Design and Embedded Systems through grading and technical mentoring.

Lecturer

Isfahan University of Medical Sciences (2013 – 2017), Sheikhbahaee University (2007 – 2012) — Isfahan, Iran

- Taught courses in computer science and health informatics, including Computer Graphics (OpenGL), C++, Multimedia Systems, and Data Storage & Retrieval.
- Supervised student projects in computer vision applications such as face detection, cell detection, and image compression using OpenCV.

Graduate Research Assistant- Microscopic Image Processing

Iranian Research Institute of Plant Protection (IRIPP) — Tehran, Iran | Apr 2006 – Jun 2008

- Developed MATLAB-based image processing tools for microscopic analysis, including fungal spore measurement, cell segmentation, and object classification using graph-based and neural network methods.

EDUCATION

Ph.D. in Computer Science, University of Nevada, Las Vegas (UNLV) | 2017 – 2022

M.S. in Computer Science, Sharif University of Technology, Iran | 2005 – 2007

B.S. in Computer Science, University of Tabriz, Iran | 2000 – 2004

SELECTED PUBLICATIONS

Journal Articles

Nasr-Esfahani, S., & Jagannathan, S. *Self-Calibrating UAV Navigation: Reinforcement Learning Approaches for Horizontal Trajectory Estimation*. IEEE Transactions on Aerospace and Electronic Systems, Early Access, 20 February 2026.

Nasr-Esfahani, S., & Jagannathan, S. *SIFT Feature-Based Relative Altitude Estimation Enhanced with a Siamese Network*. IEEE Transactions on Geoscience and Remote Sensing, 2024.

Nasr-Esfahani, S., et al. *Detection of Pits in Olive Using Hyperspectral Imaging Data*. IEEE Access, 2022.

Conference Papers

Afsharmovahed, M., **Nasr-Esfahani, S., et al.** *Automatic Detection and Quantification of Defects Toward Bridge Condition Assessment*. Transportation Research Board Annual Meeting, 2026.M.

Nasr-Esfahani, S., & Sarangapani, J. *Relative Altitude Estimation of Infrared Thermal UAV Images Using SIFT Features*. IEEE International Workshop on Multimedia Signal Processing (MMSP), 2024.

Nasr-Esfahani, S., & Sarangapani, J. *Lifelong Direct Error-Driven Learning for UAV Altitude Estimation Under Varying Weather Conditions*. IEEE International Workshop on Multimedia Signal Processing (MMSP), 2024.